INCH-POUND
MIL-DTL-24231/13E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/13D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, RECEPTACLE, MULTI-PIN, TYPE V, SEVEN- AND NINE- CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 5.

2. Cable shall be Type LS2SWU-1 in accordance with MIL-DTL-24643/33 and Type 7SS-2 and FSS-2 in accordance with MIL-DTL-915/8.

3. Specify connections, if required, for nine-pin connector.

4. All dimensions are in inches.

5. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½degree.

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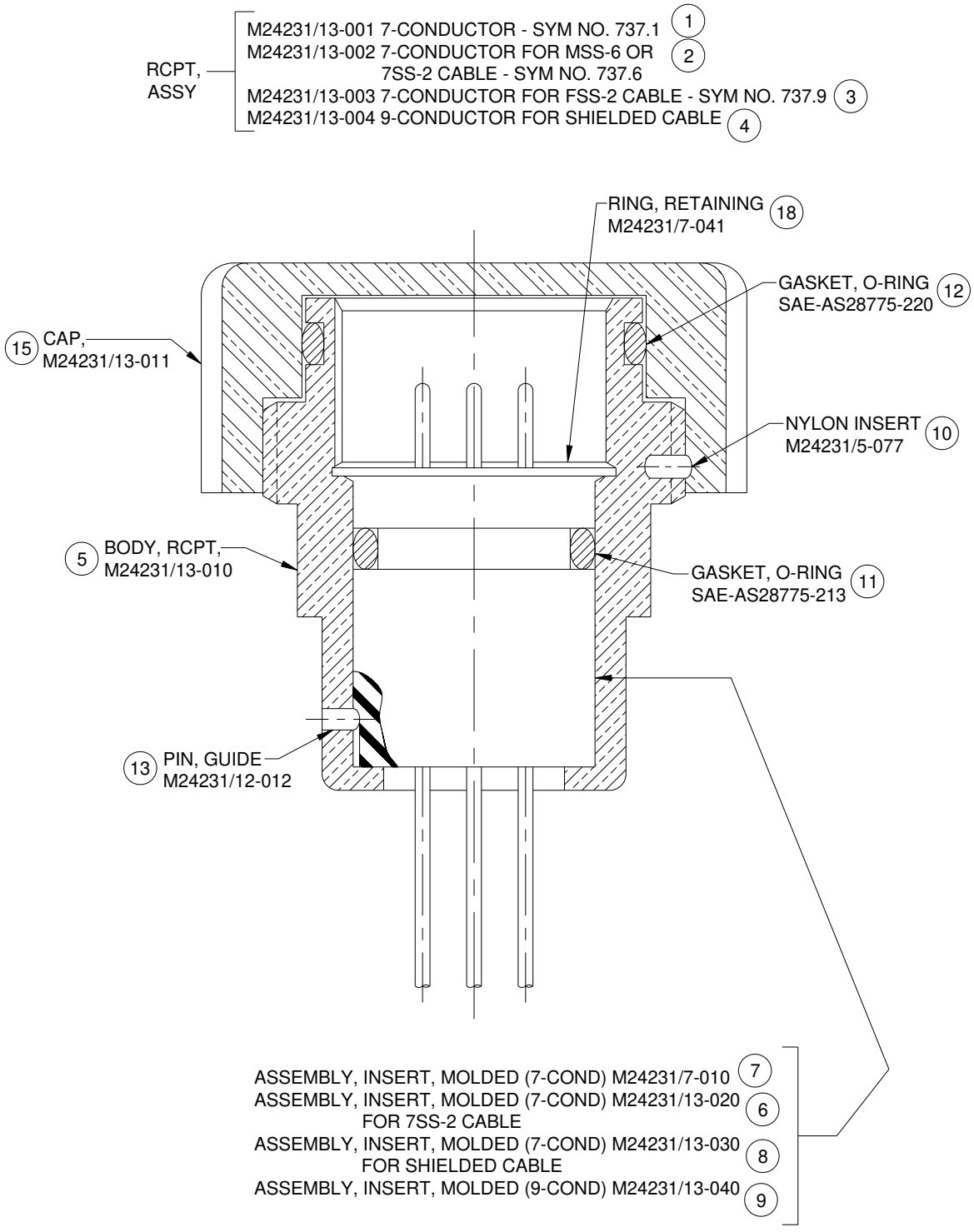


FIGURE 1. Connector, multi-pin receptacle – typical.

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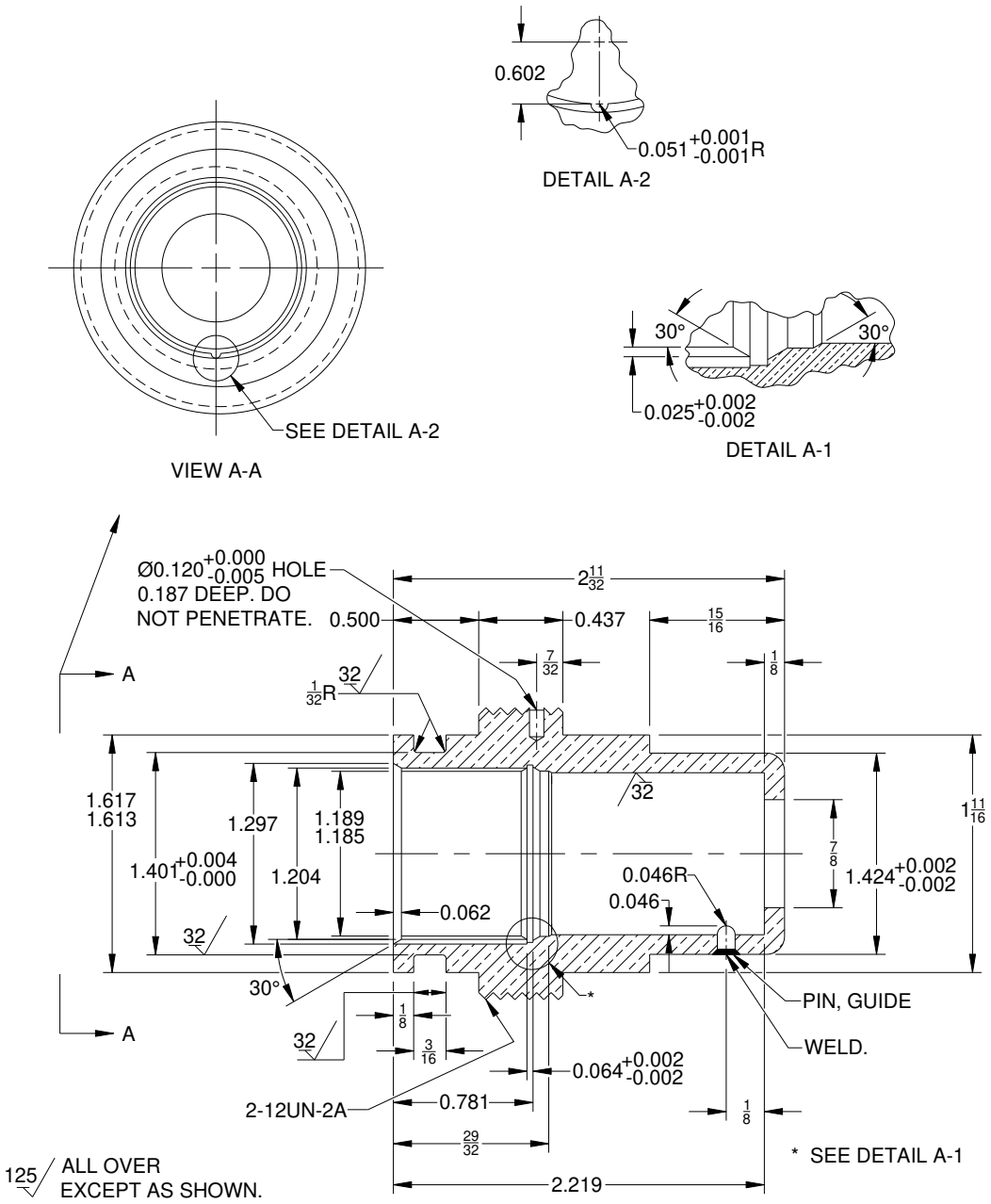


FIGURE 2. Receptacle, body – M24231/13-010. (5)

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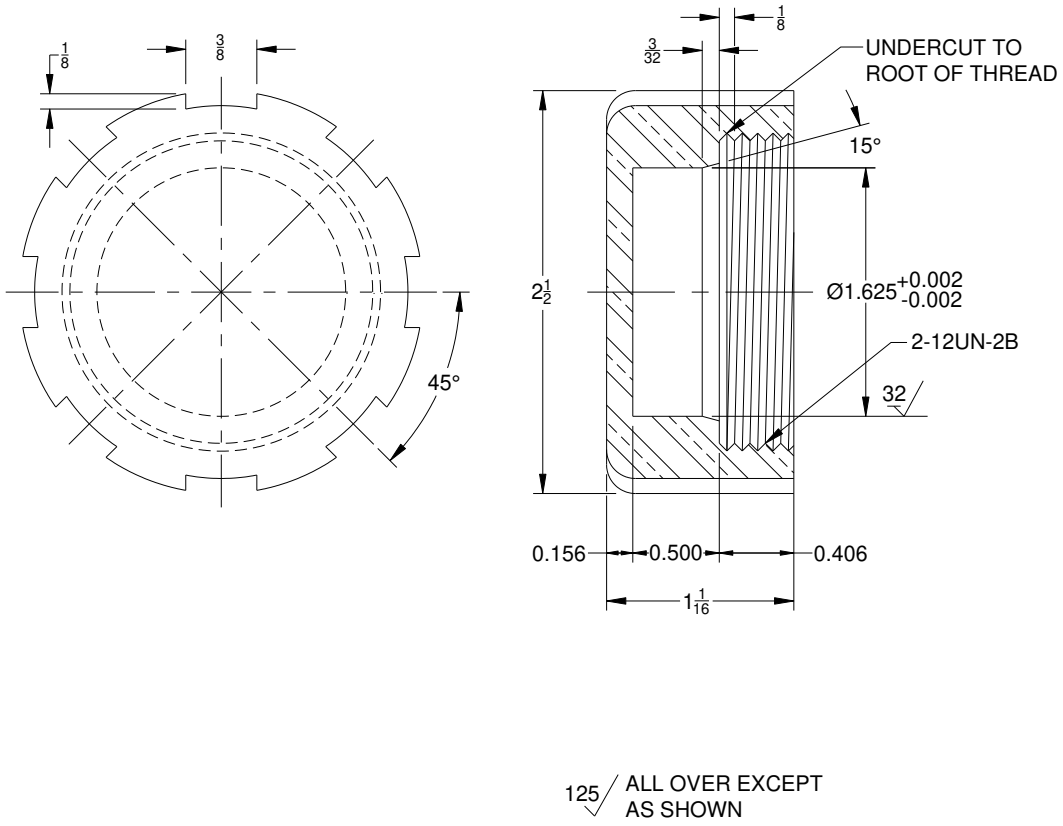


FIGURE 3. Cap – M24231/13-011. (15)

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CONNECTIONS FOR M24231/13-020		
PIN NO.	COND. COLOR	COND.
1	BLACK	SHIELDED CABLE NO. 1
2	WHITE	
7	SHIELD	
3	BLACK	SHIELDED CABLE NO. 2
4	WHITE	
7	SHIELD	
5	BLACK	(16) WIRE
6	BLACK	(16) WIRE

CONNECTIONS FOR M24231/13-030		
PIN NO.	COND. COLOR	COND.
1	BLACK	SHIELDED CABLE NO. 1
4	WHITE	
7	SHIELD	
2	BLACK	SHIELDED CABLE NO. 2
3	WHITE	
NOT CONN	SHIELD	
5	BLACK	(16) WIRE
6	BLACK	(16) WIRE

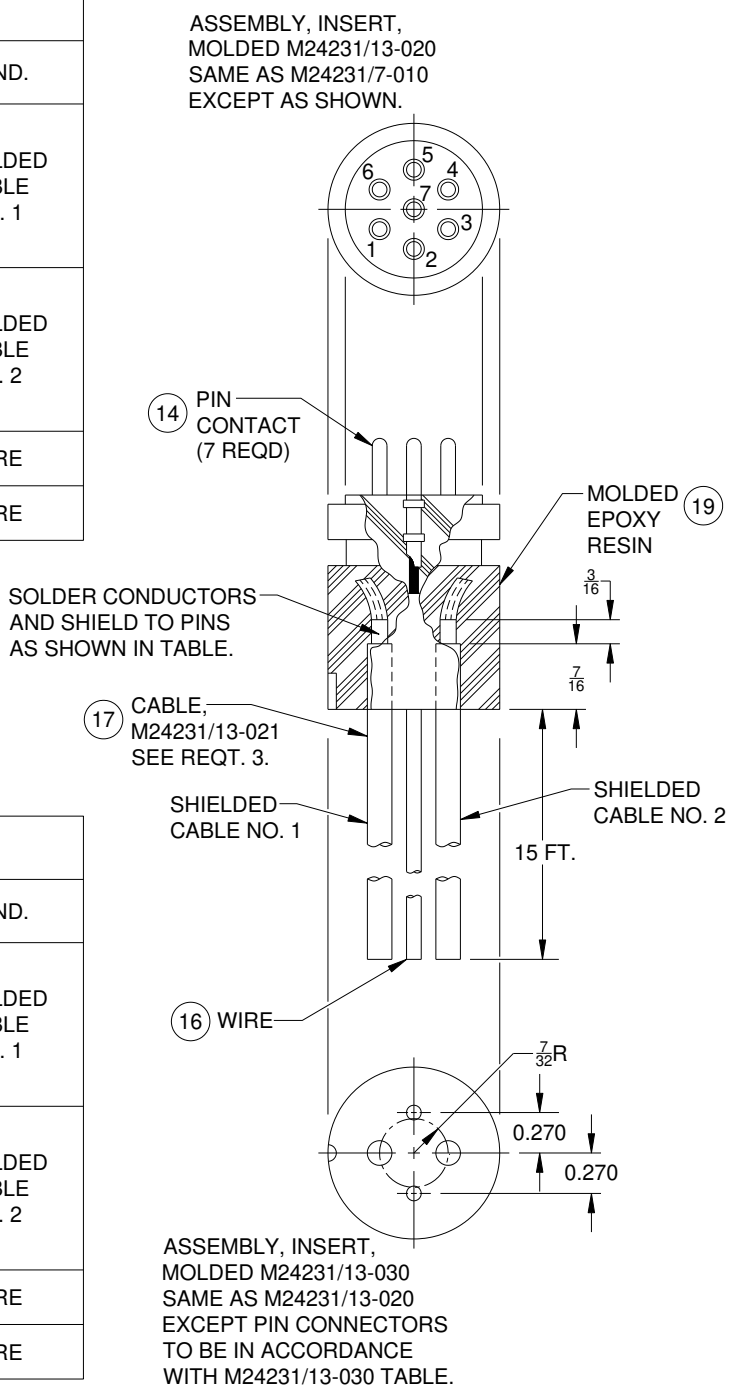
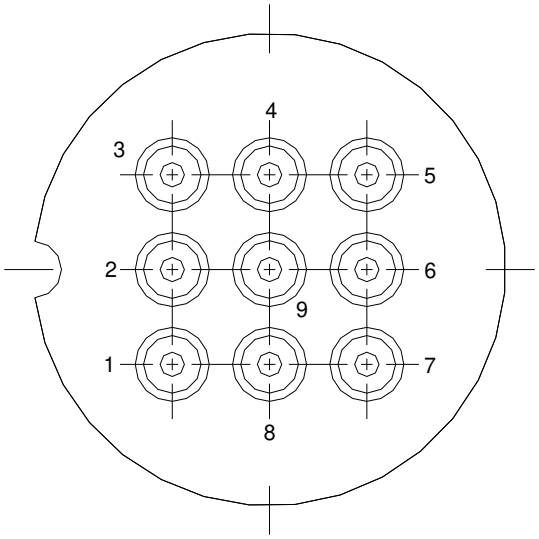


FIGURE 4. Assembly, insert, molded. (6)(7)(8)

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ASSEMBLY, INSERT, MOLDED M24231/13-040 SAME AS M24231/13-020 EXCEPT AS SHOWN. PIN CONFIGURATION SAME AS M24231/9-020. CONNECTION SHALL BE ACCORDANCE WITH ORDERING DATA.

FIGURE 5. Assembly, insert, molded. (9)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/13- 3/				Material/specification	Remarks
		001	002	003	004		
1	Connector, receptacle, 7-conductor, symbol no. 737.1	1				-----	See figure 1.
2	Connector, receptacle, 7-conductor, symbol no. 737.6		1			-----	See figure 1.
3	Connector, receptacle, 7-conductor, symbol no. 737.9			1		-----	See figure 1.
4	Connector, receptacle, 9-conductor				1	-----	See figure 1.
5	Body, receptacle	1	1	1	1	Ni-Cu, Class A, QQ-N-281	See figure 2.
6	Insert assembly, molded	1				See items 14, 16, 17, and 19.	See figure 4.
7	Insert assembly, molded		1			See items 14, 16, 17, and 19.	See figure 4.
8	Insert assembly, molded			1		See items 14, 16, 17, and 19.	See figure 4.
9	Insert assembly, molded				1	See items 14, 16, 17, and 19.	See figure 5.
10	Insert	1	1	1	1	M24231/5-077	See figure 1.
11	Gasket, O-ring	1	1	1	1	SAE-AS28775-213 4/	Lubricate with silicone compound, SAE-AS8660; see figure 1.
12	Gasket, O-ring	1	1	1	1	SAE-AS28775-220 4/	See item 11.
13	Pin, guide	1	1	1	1	M24231/12-012	See figure 1 and figure 2.
14	Pin, contact	7	7	7	9	MIL-DTL-24231	See figure 4.
15	Cap, pressure-proof	1	1	1	1	Ni-Al-BRZ, ASTM B150/B150M 5/	See figure 3.
16	Wire	As required				MIL-DTL-16878/36 BJE0	See figure 4.
17	Cable	As required				MIL-DTL-24643/33	See figure 4 and requirement 2.
18	Ring, retaining	1	1	1	1	M24231/7-041	See figure 1.
19	Epoxy resin, molded	As required				Cast epoxy resin compound 6	See figure 4.

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TABLE I. Parts list – Continued.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

5/ May be high impact plastic, PVC, commercial when pressure-proof cap is not specified (see requirement for pressure-proof caps in MIL-DTL-24231).

6/ Arco Set 200, M.M. Reif and Co., Ivyland, PA, or PR610-TCF (Chockfast Orange), ITW Philadelphia Resins Corp., Montgomeryville, PA, or equal.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-062)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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